

DEWDRIVE · REQUIREMENTS

DewDrive requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with a design point and concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

DewDrive requirements

These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs, and will be checked by calculation at TRL 3 and revised after co-design sessions. Two requirements are not met by the current concept (R11 cost, and R2 at risk), and several can only be verified by test.

■ Design point

Unless a requirement says otherwise, it applies at this design point (proposed):

- Night: 10 h adsorption at 20 °C air and 40 % RH (about 6.9 g/m³ of vapour).
- Day: about 6.0 kWh/m² of sun on the tilted aperture, with about 4.5 kWh/m² in the 6 h desorption window; air at 35 °C.
- Low-humidity case: night air at 20 °C and 25 % RH, same sun.

■ Requirements

Table 1. Requirements and concept status. "Estimate" means a first-order figure from DWD-PRC-001, to be checked at TRL 3.

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
R1	Produce water at the design point	0.5 L per day or more collected, per unit	Mass balance calculation from sorbent isotherms; later field test	Estimate about 0.49 L per day (range 0.35 to 0.6). Met at the nominal estimate, no margin
R2	Produce water in dry nights	0.25 L per day or more at 25 % night RH	Same	Estimate about 0.25 L per day. At risk: depends on the CaCl ₂ hydrate behaviour below its deliquescence point
R3	Use only the sun for regeneration	No fuel or grid power; bed reaches 75 °C or more by midday on a clear day at 35 °C ambient	Collector heat balance	Estimate 80 to 90 °C; similar glazed absorbers reached 94 °C (LaPotin et al., 2021). Met by estimate
R4	Run its own electrical loads	Fan and logger need 30 Wh per day or less, supplied by the on-board PV panel, with 2 days or more of battery autonomy	Energy budget	Estimate about 24 Wh per day against about 35 Wh per day from a 10 W panel; about 3 days autonomy. Met, thin margin
R5	Keep the condenser cool	Condenser 15 K or less above ambient at the peak of desorption	Condenser heat balance	Estimate about 12 K with a low-emissivity tray underside. Met by estimate, unverified
R6	Keep the water clean	All wetted parts food-grade; no salt carry-over (chloride in the collected water below 250 mg/L)	Materials review; later laboratory water test	Not verifiable at TRL 2. Materials chosen to meet it
R7	Contain the salt	No liquid CaCl ₂ solution leaves the trays, even after a night at 90 % RH	Salt loading calculation against uptake at 90 % RH; later chamber test	Unverified. Salt loading capped at about 33 wt % for this reason
R8	Be simple to operate	Two user actions per day (open flaps in the evening, close them in the morning), 5 min per day or less	Task analysis; co-design sessions	Met by design
R9	Be portable and quick to set up	Box 35 kg or less (two-person lift); stand separable; set up by two people in 30 min or less	Mass estimate; later trial	Estimate box about 30 kg, total about 45 kg. Met by estimate
R10	Survive the site	Stable in 20 m/s (72 km/h) wind with anchors or ballast; UV-stable glazing; sorbent 300 cycles or more with 20 % or less loss of capacity	Wind load calculation; supplier data; later cycling test	Unverified. Wind uplift estimate about 330 N, so anchors or about 40 kg of ballast are needed
R11	Cost	Parts \$400 or less (<code>project.yaml</code> budget)	Priced BOM	Not met: about \$475 (indicative), about 19 % over
R12	Record performance	Log air temperature and RH, bed and condenser temperature every 5 min for 30 days or more; daily water mass recorded by weighing	Design review	Met by design (logger and sensors, items 12 and 13)

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
R13	Protect users	External surfaces a person would touch in normal use stay at 60 °C or less; the glazing is labelled hot; no exposed conductor above 60 V DC	Surface temperature estimate; design review	Met by design at 12.8 V; glazing surface temperature unverified

■ Assumptions

- The composite (about 33 wt % CaCl_2 in mesoporous silica gel) takes up about 0.30 g/g at 40 % RH and 20 °C and about 0.16 g/g at 25 % RH. These are estimates between published values for similar composites (0.52 g/g at 60 % RH for a LiCl and silica gel composite, Shao et al., 2024; up to 0.6 to 0.7 g/g for SWS-1L, Aristov et al., 2006). They must be replaced by measured isotherms at TRL 3.
- Half of the overnight uptake is released by day. That needs the bed near 85 °C with the condenser at 45 to 50 °C.
- One unit is a supplement: about 0.5 L per day is about one fifth of one person's survival drinking need of 2.5 to 3 L per day (WHO technical note 9).

Safety: Requirements R6, R7, R10 and R13 exist because of hazards: untreated water, a corrosive salt solution, wind loads on a 1.1 m² panel, and hot surfaces. They cannot be waived without a documented decision by Amish.